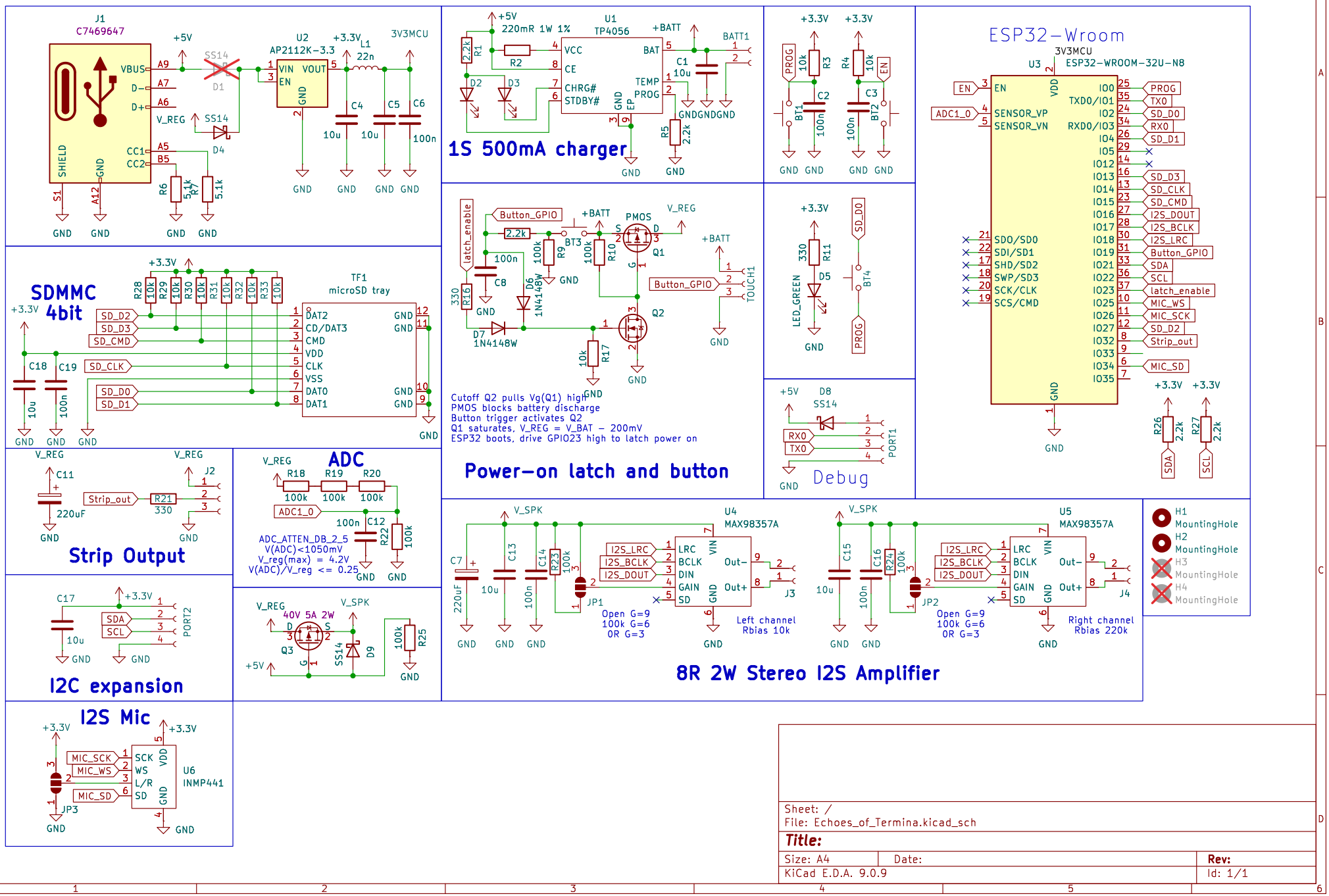

J1
C7469647
VBUS
D-
D+
CC1
CC2
SHIELD
GND
S1
A12
+5V
V_REG
SS14
D1
D4
R6 5.1k
R7 5.1k
U2
AP2112K-3.3
VIN
VOUT
EN
GND
+3.3V
L1
22n
3V3MCU
C4 10u
C5 10u
C6 100n
SDMMC
4bit
TF1
microSD tray
SD_D2
SD_D3
SD_CMD
SD_CLK
SD_D0
SD_D1
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
C18 10u
C19 100n
R28 10k
R29 10k
R30 10k
R31 10k
R32 10k
R33 10k
Strip Output
C11 220uF
Strip_out
R21 330
J2
ADC
R18 100k
R19 100k
R20 100k
ADC1_0
C12 100n
R22 100k
ADC_ATTEN_DB_2_5
V(ADC)<1050mV
V_reg(max) = 4.2V
V(ADC)/V_reg <= 0.25
I2C expansion
C17 10u
SDA
SCL
PORT2
I2S Mic
U6
INMP441
MIC_SCK
MIC_WS
MIC_SD
SCK
WS
L/R
SD
VDD
JP3
1S 500mA charger
U1
TP4056
+5V
220mR 1W 1%
R1 2.2k
R2
D2
D3
VCC
CE
CHRG#
STDBY#
BAT
TEMP
PROG
EP
+BATT
BATT1
C1 10u
R5 2.2k
PROG 10k
R3
C2 100n
BT1
R4 10k
EN
C3 100n
BT2
Power-on latch and button
latch_enable
Button_GPIO
2.2k
R9 100k
BT3
PMOS
Q1
R10 100k
C8 100n
R16 330
D6 1N4148W
D7 1N4148W
R17 10k
Q2
TOUCH1
Cutoff Q2 pulls Vg(Q1) high
PMOS blocks battery discharge
Button trigger activates Q2
Q1 saturates, V_REG = V_BAT - 200mV
ESP32 boots, drive GPIO23 high to latch power on
R11 330
D5
LED_GREEN
SD_D0
BT4
PROG
D8
SS14
RX0
TX0
PORT1
Debug
ESP32-Wroom
U3
ESP32-WROOM-32U-N8
EN
ADC1_0
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
PROG
TX0
SD_D0
RX0
SD_D1
SD_D3
SD_CLK
SD_CMD
I2S_DOUT
I2S_BCLK
I2S_LRC
Button_GPIO
SDA
SCL
latch_enable
MIC_WS
MIC_SCK
SD_D2
Strip_out
MIC_SD
R26 2.2k
R27 2.2k
8R 2W Stereo I2S Amplifier
V_SPK
U4
MAX98357A
U5
MAX98357A
C7 220uF
C13 10u
C14 100n
R23 100k
JP1
Open G=9
100k G=6
OR G=3
LRC
BCLK
DIN
GAIN
SD
VIN
Out-
Out+
GND
J3
Left channel
Rbias 10k
C15 10u
C16 100n
R24 100k
JP2
J4
Right channel
Rbias 220k
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
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KiCad E.D.A. 9.0.9
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